1.49 Proof by Induction: Sequences

Worked Solutions

1 a)

$$3\left(3^{k-1}+\frac{1}{2}\right)-1=3\times 3^{k-1}+\frac{3}{2}-1=3^1\times 3^{k-1}+\frac{1}{2}=3^k+\frac{1}{2} \text{ as req.}$$

1 b)

$$3(2^k+3^k)-2^k=3\times 2^k+3\times 3^k-2^k=2^1\times 2^k+3^1\times 3^k=2^{k+1}+3^{k+1} \text{ as req.}$$

1 c)

$$\frac{\frac{3}{3k-1}}{1+\frac{3}{3k-1}}=\frac{3}{3k-1+3}=\frac{3}{3k+2}=\frac{3}{3(k+1)-1} \text{ as req.}$$

1 d)

$$\begin{aligned}\frac{1}{2}(k^2-5k+12)+k-2&=\frac{1}{2}(k^2-5k+12+2k-4)\\&=\frac{1}{2}(k^2-3k+8)=\frac{1}{2}((k+1)^2-5k+7)\\&=\frac{1}{2}((k+1)^2-5(k+1)+12) \text{ as req.}\end{aligned}$$

2)

Prove true for $n=1$: $u_1=11-3\times 1=8$ so true for $n=1$.

Assume true for $n=k$: $u_k=11-3k$.

Prove true for $n=k+1$:

$$u_{k+1}=u_k-3=11-3k-3=11-3(k+1),$$

which is $u_n=11-3n$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq 1$.

3)

Prove true for $n=1$: $u_1=2^{1-1}+1=1+1=2$ so true for $n=1$.

Assume true for $n=k$: $u_k=2^{k-1}+1$.

Prove true for $n=k+1$:

$$\begin{aligned}u_{k+1}&=2u_k-1=2(2^{k-1}+1)-1=2\times 2^{k-1}+2-1\\&=2^1\times 2^{k-1}+2-1=2^k+1=2^{(k+1)-1}+1,\end{aligned}$$

which is $u_n=2^{n-1}+1$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq 1$.

4)

Prove true for $n = 1$: $u_1 = 9 \times 4^{1-1} - 1 = 9 - 1 = 8$ so true for $n = 1$.

Assume true for $n = k$: $u_k = 9 \times 4^{k-1} - 1$.

Prove true for $n = k + 1$:

$$u_{k+1} = 4u_k + 3 = 4(9 \times 4^{k-1} - 1) + 3$$
$$= 9 \times 4^1 \times 4^{k-1} - 4 + 3 = 9 \times 4^k - 1 = 9 \times 4^{(k+1)-1} - 1,$$

which is $u_n = 9 \times 4^{n-1} - 1$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

5)

Prove true for $n = 1$: $u_1 = \frac{1}{6}(7 \times 3^1 + 3) = \frac{1}{6}(21 + 3) = 4$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{1}{6}(7 \times 3^k + 3)$.

Prove true for $n = k + 1$:

$$u_{k+1} = 3u_k - 1 = 3\left[\frac{1}{6}(7 \times 3^k + 3)\right] - 1$$
$$= \frac{1}{6}(7 \times 3^1 \times 3^k + 9) - 1 = \frac{1}{6}(7 \times 3^1 \times 3^k + 9 - 6)$$
$$= \frac{1}{6}(7 \times 3^{k+1} + 3),$$

which is $u_n = \frac{1}{6}(7 \times 3^n + 3)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

6)

Prove true for $n = 1$: $u_1 = \frac{19 \times 8 - 40}{56} = \frac{112}{56} = 2$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{19 \times 8^k - 40}{56}$.

Prove true for $n = k + 1$:

$$u_{k+1} = 8u_k + 5 = 8\left(\frac{19 \times 8^k - 40}{56}\right) + 5$$
$$= \frac{19 \times 8 \times 8^k - 320}{56} + \frac{280}{56} = \frac{19 \times 8^1 \times 8^k - 40}{56}$$
$$= \frac{19 \times 8^{k+1} - 40}{56},$$

which is $u_n = \frac{19 \times 8^n - 40}{56}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

7)

Prove true for $n = 1$: $u_1 = \frac{1}{2}(3 + 7 - 6) = \frac{1}{2} \times 4 = 2$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{1}{2}(3k^2 + 7k - 6)$.

Prove true for $n = k + 1$:

$$u_{k+1} = u_k + 3k + 5 = \frac{1}{2}(3k^2 + 7k - 6) + 3k + 5$$

$$= \frac{1}{2}(3k^2 + 7k - 6 + 6k + 10) = \frac{1}{2}(3k^2 + 13k + 4)$$

$$= \frac{1}{2}(3(k+1)^2 + 7k + 1) = \frac{1}{2}(3(k+1)^2 + 7(k+1) - 6),$$

which is $u_n = \frac{1}{2}(3n^2 + 7n - 6)$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

8)

Prove true for $n = 1$: $u_1 = \frac{1}{1} = 1$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{1}{k}$.

Prove true for $n = k + 1$:

$$u_{k+1} = \frac{u_k}{1 + u_k} = \frac{\frac{1}{k}}{1 + \frac{1}{k}} = \frac{1}{k+1},$$

which is $u_n = \frac{1}{n}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9)

Prove true for $n = 1$: $u_1 = \frac{3^1+1}{3^1-1} = \frac{4}{2} = 2$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{3^k+1}{3^k-1}$.

Prove true for $n = k + 1$:

$$u_{k+1} = \frac{2u_k + 1}{u_k + 2} = \frac{2(\frac{3^k+1}{3^k-1}) + 1}{(\frac{3^k+1}{3^k-1}) + 2}$$

$$= \frac{2(3^k + 1) + 3^k - 1}{3^k + 1 + 2(3^k - 1)} = \frac{2 \times 3^k + 2 + 3^k - 1}{3^k + 1 + 2 \times 3^k - 2}$$

$$= \frac{3 \times 3^k + 1}{3 \times 3^k - 1} = \frac{3^{k+1} + 1}{3^{k+1} - 1},$$

which is $u_n = \frac{3^n+1}{3^n-1}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

10 a)

$$z_2 = (1+i)z_1 + (1-i)^1 = (1+i)(1+i) + (1-i) = 1 + 2i - 1 + 1 - i = 1 + i.$$

$$z_3 = (1+i)z_2 + (1-i)^2 = (1+i)(1+i) + (1-i)^2 = 2i + (1 - 2i - 1) = 0.$$

10 b)

Prove true for $n = 1$:

$$z_1 = \frac{1}{2}(1+i)^1 + \frac{i}{2}(1-i)^1 = \frac{1}{2} + \frac{1}{2}i + \frac{1}{2}i + \frac{1}{2} = 1 + i,$$

so true for $n = 1$.

Assume true for $n = k$: $z_k = \frac{1}{2}(1+i)^k + \frac{i}{2}(1-i)^k$.

Prove true for $n = k + 1$:

$$\begin{aligned} z_{k+1} &= (1+i)z_k + (1-i)^k \\ &= (1+i)\left[\frac{1}{2}(1+i)^k + \frac{i}{2}(1-i)^k\right] + (1-i)^k \\ &= \frac{1}{2}(1+i)^{k+1} + \frac{i}{2}(1+i)(1-i)^k + (1-i)^k \\ &= \frac{1}{2}(1+i)^{k+1} + \frac{i}{2}(1-i)^k[(1+i) + \frac{2}{i}] \\ &= \frac{1}{2}(1+i)^{k+1} + \frac{i}{2}(1-i)^k[1-i] \\ &= \frac{1}{2}(1+i)^{k+1} + \frac{i}{2}(1-i)^{k+1}, \end{aligned}$$

which is $z_n = \frac{1}{2}(1+i)^n + \frac{i}{2}(1-i)^n$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

11 a)

$$\begin{aligned} u_2 &= \frac{u_1}{2+u_1} = \frac{3}{2+3} = \frac{3}{5} \\ u_3 &= \frac{u_2}{2+u_2} = \frac{\frac{3}{5}}{2+\frac{3}{5}} = \frac{3}{13} \\ u_4 &= \frac{u_3}{2+u_3} = \frac{\frac{3}{13}}{2+\frac{3}{13}} = \frac{3}{29}. \end{aligned}$$

11 b)

$$2^{1+1} - 3 = 1 \qquad 2^{2+1} - 3 = 5 \qquad 2^{3+1} - 3 = 13 \qquad 2^{4+1} - 3 = 29.$$

11 c)

$$u_n = \frac{3}{2^{n+1} - 3}.$$

11 d)

Prove true for $n = 1$: $u_1 = \frac{3}{2^{1+1}-3} = \frac{3}{4-3} = 3$ so true for $n = 1$.

Assume true for $n = k$: $u_k = \frac{3}{2^{k+1}-3}$.

Prove true for $n = k + 1$:

$$\begin{aligned} u_{k+1} &= \frac{u_k}{2+u_k} = \frac{\frac{3}{2^{k+1}-3}}{2+\frac{3}{2^{k+1}-3}} \\ &= \frac{3}{2(2^{k+1}-3)+3} = \frac{3}{2 \times 2^{k+1} - 6 + 3} \\ &= \frac{3}{2^1 \times 2^{k+1} - 3} = \frac{3}{2^{k+2} - 3} = \frac{3}{2^{(k+1)+1} - 3}, \end{aligned}$$

which is $u_n = \frac{3}{2^{k+1}-3}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

12)

Prove true for $n = 1$: $u_1 = q + (1-1)a = q$ so true for $n = 1$.

Assume true for $n = k$: $u_k = q + (k-1)a$.

Prove true for $n = k + 1$:

$$u_{k+1} = u_k + a = q + (k-1)a + a = q + ka$$

$$= q + (k + 1 - 1)a,$$

which is $u_n = q + (n-1)a$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

13 a)

$$u_3 = 3u_2 - 2u_1 = 3 \times 13 - 2 \times 5 = 29.$$

$$u_4 = 3u_3 - 2u_2 = 3 \times 29 - 2 \times 13 = 61.$$

13 b)

Prove true for $n = 1$ and $n = 2$:

$u_1 = -3 + 2^{1+2} = -3 + 8 = 5$ so true for $n = 1$, $\qquad u_2 = -3 + 2^{2+2} = 13$ so true for $n = 2$.

Assume true for $n = k$: $u_k = -3 + 2^{k+2}$.

Assume true for $n = k + 1$: $u_{k+1} = -3 + 2^{k+3}$.

Prove true for $n = k + 2$:

$$u_{k+2} = 3u_{k+1} - 2u_k = 3(-3 + 2^{k+3}) - 2(-3 + 2^{k+2})$$

$$= -9 + 3 \times 2^{k+3} + 6 - 2 \times 2^{k+2}$$

$$= -3 + 6 \times 2^{k+2} - 2 \times 2^{k+2} = -3 + 4 \times 2^{k+2}$$

$$= -3 + 2^{k+4} = -3 + 2^{(k+2)+2},$$

which is $u_n = -3 + 2^{n+2}$ with $k + 2$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.